

Unit 8. The Spectral Theorem

1 Orthogonal diagonalisation

1.1 The real spectral theorem

Definition 8.1 (Orthogonal diagonalisation). A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **orthogonally diagonalisable** if there are an orthogonal $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and a real diagonal $\mathbf{\Lambda}$ such that

$$\mathbf{Q}'\mathbf{A}\mathbf{Q} = \mathbf{\Lambda}, \quad \mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}'.$$

If $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_n)$ and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$, then

$$\mathbf{A}\mathbf{q}_i = \lambda_i\mathbf{q}_i, \quad \mathbf{q}_i'\mathbf{q}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Origin: H §21.5a and Thm. 21.5.1; A&M Ex. 7.46

Theorem 8.2 (Real eigenpair). Every real symmetric matrix of positive order has a real eigenvalue and a real unit eigenvector.

Origin: H Thms. 21.5.5–21.5.6; A&M Exs. 7.40 and 7.43

Proof. Let $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$, $n \geq 1$. The complex eigenvalue theory gives $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$ for some $\lambda \in \mathbb{C}$ and $\mathbf{0} \neq \mathbf{z} \in \mathbb{C}^n$. Since $\mathbf{A}^* = \mathbf{A}$,

$$\overline{\mathbf{z}^* \mathbf{A} \mathbf{z}} = \mathbf{z}^* \mathbf{A}^* \mathbf{z} = \mathbf{z}^* \mathbf{A} \mathbf{z}, \quad \mathbf{z}^* \mathbf{z} > 0.$$

Thus $\mathbf{z}^* \mathbf{A} \mathbf{z} = \lambda \mathbf{z}^* \mathbf{z}$ is real and $\lambda \in \mathbb{R}$. Write $\mathbf{z} = \mathbf{u} + i\mathbf{v}$ with $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Then

$$\mathbf{A}\mathbf{u} + i\mathbf{A}\mathbf{v} = \lambda\mathbf{u} + i\lambda\mathbf{v} \implies \mathbf{A}\mathbf{u} = \lambda\mathbf{u}, \quad \mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

At least one of $\mathbf{u}, \mathbf{v}$ is nonzero; normalising that vector gives the assertion. □

Theorem 8.3 (Real spectral theorem). For $\mathbf{A} \in \mathbb{R}^{n \times n}$,

$$\mathbf{A} = \mathbf{A}' \iff \mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}'$$

for some orthogonal $\mathbf{Q}$ and real diagonal $\mathbf{\Lambda}$.

Origin: H Thm. 21.5.7; A&M Ex. 7.46; S Cor. 5.3

Proof. Suppose first that $\mathbf{A} = \mathbf{A}'$. Induct on n . The assertion holds for $n = 1$. For $n \geq 2$, Theorem 8.2 gives

$$\mathbf{A}\mathbf{u} = \lambda\mathbf{u}, \quad \mathbf{u}'\mathbf{u} = 1.$$

Complete $\mathbf{u}$ to an orthonormal basis and write $\mathbf{Q}_0 = (\mathbf{u}, \mathbf{V})$. Symmetry gives

$$\mathbf{Q}_0' \mathbf{A} \mathbf{Q}_0 = \begin{pmatrix} \mathbf{u}' \mathbf{A} \mathbf{u} & \mathbf{u}' \mathbf{A} \mathbf{V} \\ \mathbf{V}' \mathbf{A} \mathbf{u} & \mathbf{V}' \mathbf{A} \mathbf{V} \end{pmatrix} = \begin{pmatrix} \lambda & \mathbf{0}' \\ \mathbf{0} & \mathbf{C} \end{pmatrix}, \quad \mathbf{C}' = \mathbf{C}.$$

By induction, $\mathbf{C} = \mathbf{R} \mathbf{D} \mathbf{R}'$ for an orthogonal $\mathbf{R}$. Hence

$$\mathbf{Q} = \mathbf{Q}_0 \operatorname{diag}(1, \mathbf{R}), \quad \mathbf{Q}' \mathbf{A} \mathbf{Q} = \operatorname{diag}(\lambda, \mathbf{D}).$$

Conversely, if $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}'$ with $\mathbf{Q}' \mathbf{Q} = \mathbf{I}$ and $\mathbf{\Lambda}' = \mathbf{\Lambda}$, then

$$\mathbf{A}' = \mathbf{Q} \mathbf{\Lambda}' \mathbf{Q}' = \mathbf{A}.$$

□

Corollary 8.4 (Orthogonal eigenspace decomposition). Let $\theta_1, \dots, \theta_s$ be the distinct eigenvalues of a real symmetric $\mathbf{A}$. Then

$$\mathbb{R}^n = \mathcal{E}_{\theta_1}(\mathbf{A}) \perp \oplus \cdots \perp \oplus \mathcal{E}_{\theta_s}(\mathbf{A}).$$

For every j , the geometric and algebraic multiplicities of θ_j are equal.

Origin: H Cor. 21.5.8; A&M Exs. 7.47–7.48

Proof. Choose the columns of $\mathbf{Q}$ in Theorem 8.3 in blocks $\mathbf{Q} = (\mathbf{Q}_1, \dots, \mathbf{Q}_s)$ corresponding to the distinct diagonal values. Then

$$\mathbf{A}\mathbf{Q}_j = \theta_j\mathbf{Q}_j, \quad \mathbf{Q}'_i\mathbf{Q}_j = \begin{cases} \mathbf{I}, & i = j, \\ \mathbf{0}, & i \neq j. \end{cases}$$

The columns of $\mathbf{Q}$ form a basis, so their block column spaces give the orthogonal direct sum. Moreover,

$$\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{Q}'\mathbf{A}\mathbf{Q}) = \prod_{j=1}^s (t - \theta_j)^{\dim \mathcal{E}_{\theta_j}(\mathbf{A})}.$$

The exponent is both multiplicities. □

Example 8.1. For

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 & 2^{-1/2} & 2^{-1/2} \\ 0 & 2^{-1/2} & -2^{-1/2} \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{\Lambda} = \text{diag}(5, 3, 1),$$

direct multiplication gives

$$\mathbf{Q}'\mathbf{Q} = \mathbf{I}_3, \quad \mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}, \quad \mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}'.$$

Origin: Adapted from A&M Exs. 7.44–7.46

1.2 Spectral projectors

Definition 8.5 (Spectral projector). Let $\mathbf{A} = \mathbf{A}'$ have distinct eigenvalues $\theta_1, \dots, \theta_s$, and let the columns of $\mathbf{Q}_j$ be an orthonormal basis of $\mathcal{E}_{\theta_j}(\mathbf{A})$. The **spectral projector**

for θ_j is

$$\mathbf{P}_j = \mathbf{Q}_j \mathbf{Q}_j'.$$

Origin: H eqs. (21.5.4)–(21.5.5)

Theorem 8.6 (Spectral projector algebra). With Definition 8.5,

$$\mathbf{P}_j' = \mathbf{P}_j, \quad \mathbf{P}_i \mathbf{P}_j = \begin{cases} \mathbf{P}_j, & i = j, \\ \mathbf{O}, & i \neq j, \end{cases} \quad \sum_{j=1}^s \mathbf{P}_j = \mathbf{I},$$

$$\operatorname{im} \mathbf{P}_j = \mathcal{E}_{\theta_j}(\mathbf{A}), \quad \mathbf{A} \mathbf{P}_j = \theta_j \mathbf{P}_j, \quad \mathbf{A} = \sum_{j=1}^s \theta_j \mathbf{P}_j.$$

Moreover,

$$\mathbf{P}_j = \ell_j(\mathbf{A}), \quad \ell_j(t) = \prod_{\substack{1 \leq i \leq s \\ i \neq j}} \frac{t - \theta_i}{\theta_j - \theta_i},$$

where an empty product is 1. Thus $\mathbf{P}_j$ is independent of the chosen orthonormal basis of its eigenspace.

Origin: H Ex. 21.18 and eq. (21.5.5); S §5.5

Proof. The block orthogonality in Corollary 8.4 gives

$$\mathbf{P}_i \mathbf{P}_j = \mathbf{Q}_i (\mathbf{Q}'_i \mathbf{Q}_j) \mathbf{Q}'_j, \quad \sum_j \mathbf{P}_j = \mathbf{Q} \mathbf{Q}' = \mathbf{I}.$$

Also

$$\mathbf{P}_j \mathbf{Q}_i = \delta_{ij} \mathbf{Q}_i, \quad \mathbf{A} \mathbf{P}_j = \mathbf{A} \mathbf{Q}_j \mathbf{Q}'_j = \theta_j \mathbf{P}_j.$$

These identities give the range assertion and

$$\mathbf{A} = \mathbf{A} \sum_j \mathbf{P}_j = \sum_j \theta_j \mathbf{P}_j.$$

Since $\ell_j(\theta_i) = \delta_{ij}$,

$$\ell_j(\mathbf{A}) \mathbf{Q}_i = \ell_j(\theta_i) \mathbf{Q}_i = \mathbf{P}_j \mathbf{Q}_i.$$

The columns of $\mathbf{Q}$ form a basis, so $\ell_j(\mathbf{A}) = \mathbf{P}_j$.

□

Corollary 8.7 (Rank, trace, and determinant). If $m_j = \dim \mathcal{E}_{\theta_j}(\mathbf{A})$, then

$$\text{rank } \mathbf{A} = \sum_{\theta_j \neq 0} m_j, \quad \text{tr } \mathbf{A} = \sum_{j=1}^s m_j \theta_j, \quad \det \mathbf{A} = \prod_{j=1}^s \theta_j^{m_j}.$$

Origin: H Thm. 21.6.1; A&M Ex. 7.49

Proof. Orthogonal similarity preserves rank, trace, and determinant. Apply this to

$$\mathbf{Q}'\mathbf{A}\mathbf{Q} = \text{diag}(\theta_1 \mathbf{I}_{m_1}, \dots, \theta_s \mathbf{I}_{m_s}).$$

□

1.3 Finite spectral calculus

Definition 8.8 (Function of a symmetric matrix). Let $\mathbf{A} = \mathbf{A}'$ have distinct eigenvalues $\theta_1, \dots, \theta_s$ and spectral projectors $\mathbf{P}_1, \dots, \mathbf{P}_s$. For a function $f : \{\theta_1, \dots, \theta_s\} \rightarrow \mathbb{R}$, define

$$f(\mathbf{A}) = \sum_{j=1}^s f(\theta_j) \mathbf{P}_j.$$

Origin: S §5.5; H eqs. (21.5.3)–(21.5.5)

Theorem 8.9 (Rules of finite spectral calculus). Let f, g be real functions on the spectrum of $\mathbf{A} = \mathbf{A}'$. Write its distinct eigenvalues as $\theta_1, \dots, \theta_s$, set $m_j = \dim \mathcal{E}_{\theta_j}(\mathbf{A})$, and let $a, b \in \mathbb{R}$.

$$(af + bg)(\mathbf{A}) = af(\mathbf{A}) + bg(\mathbf{A}), \quad (fg)(\mathbf{A}) = f(\mathbf{A})g(\mathbf{A}),$$

$$f(\mathbf{A})' = f(\mathbf{A}), \quad \mathbf{A}f(\mathbf{A}) = f(\mathbf{A})\mathbf{A}.$$

If $p(t) = \sum_{k=0}^d p_k t^k$ is a polynomial, this definition agrees with $p(\mathbf{A}) = \sum_{k=0}^d p_k \mathbf{A}^k$. The eigen-

values of $f(\mathbf{A})$ are $f(\theta_j)$ with multiplicities m_j . Consequently,

$$\operatorname{rank} f(\mathbf{A}) = \sum_{f(\theta_j) \neq 0} m_j, \quad \operatorname{tr} f(\mathbf{A}) = \sum_j m_j f(\theta_j), \quad \det f(\mathbf{A}) = \prod_j f(\theta_j)^{m_j}.$$

If every $f(\theta_j) \neq 0$, then

$$f(\mathbf{A})^{-1} = h(\mathbf{A}), \quad h(\theta_j) = \frac{1}{f(\theta_j)}.$$

Origin: Adapted from S Prop. 5.7; H Thm. 21.6.1

Proof. Theorem 8.6 gives

$$\mathbf{P}_i \mathbf{P}_j = \delta_{ij} \mathbf{P}_j, \quad \mathbf{A} \mathbf{P}_j = \theta_j \mathbf{P}_j.$$

Hence

$$f(\mathbf{A})g(\mathbf{A}) = \sum_{i,j} f(\theta_i)g(\theta_j) \mathbf{P}_i \mathbf{P}_j = \sum_j f(\theta_j)g(\theta_j) \mathbf{P}_j.$$

Linearity, symmetry, and commutation follow from the same identities. For $k \geq 0$,

$$\mathbf{A}^k = \left(\sum_j \theta_j \mathbf{P}_j \right)^k = \sum_j \theta_j^k \mathbf{P}_j,$$

so polynomial evaluation agrees. In the orthonormal eigenbasis,

$$\mathbf{Q}' f(\mathbf{A}) \mathbf{Q} = \text{diag}(f(\theta_1) \mathbf{I}_{m_1}, \dots, f(\theta_s) \mathbf{I}_{m_s}).$$

This gives the eigenvalue, rank, trace, and determinant formulas. If all values are nonzero, then

$$f(\mathbf{A})h(\mathbf{A}) = (fh)(\mathbf{A}) = \sum_j \mathbf{P}_j = \mathbf{I}.$$

□

Example 8.2. For $\mathbf{A}$ in Example 8.1, the projector onto $\mathcal{E}_3(\mathbf{A})$ is

$$\mathbf{P}_3 = -\frac{1}{4}(\mathbf{A} - 5\mathbf{I})(\mathbf{A} - \mathbf{I}) = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Indeed, $p(t) = -(t - 5)(t - 1)/4$ satisfies $p(5) = 0$, $p(3) = 1$, and $p(1) = 0$.

Origin: Adapted from S §5.5 and H Ex. 21.18

2 Variational characterisations

2.1 Rayleigh quotients

Definition 8.10 (Rayleigh quotient). For $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$ and $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n$, the **Rayleigh quotient** is

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{x}}.$$

Write the eigenvalues of $\mathbf{A}$, counted with multiplicity, as

$$\lambda_1(\mathbf{A}) \geq \lambda_2(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A}).$$

The argument $\mathbf{A}$ is omitted when fixed.

Origin: H Thm. 21.5.6; A&M Exs. 7.53 and 12.39

Theorem 8.11 (Rayleigh expansion and equality cases). Let $\mathbf{q}_1, \dots, \mathbf{q}_n$ be an orthonormal eigen-

basis ordered with $\lambda_1 \geq \dots \geq \lambda_n$. If $\mathbf{x} = \sum_i c_i \mathbf{q}_i \neq \mathbf{0}$, then

$$R_{\mathbf{A}}(\mathbf{x}) = \sum_{i=1}^n \lambda_i w_i, \quad w_i = \frac{c_i^2}{\sum_j c_j^2}, \quad w_i \geq 0, \quad \sum_i w_i = 1.$$

Therefore

$$\lambda_n \leq R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_1,$$

with

$$R_{\mathbf{A}}(\mathbf{x}) = \lambda_1 \iff \mathbf{x} \in \mathcal{E}_{\lambda_1}(\mathbf{A}), \quad R_{\mathbf{A}}(\mathbf{x}) = \lambda_n \iff \mathbf{x} \in \mathcal{E}_{\lambda_n}(\mathbf{A}).$$

Origin: H Thm. 21.5.6; A&M Ex. 12.39; S eqs. (6.1)–(6.2)

Proof. Orthogonality and $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$ give

$$\mathbf{x}'\mathbf{x} = \sum_i c_i^2, \quad \mathbf{x}'\mathbf{A}\mathbf{x} = \sum_i \lambda_i c_i^2.$$

Thus $R_{\mathbf{A}}(\mathbf{x})$ is the displayed weighted mean. Moreover,

$$\lambda_1 - R_{\mathbf{A}}(\mathbf{x}) = \sum_i (\lambda_1 - \lambda_i) w_i.$$

This is zero exactly when $w_i = 0$ for every i with $\lambda_i < \lambda_1$, which is equivalent to $\mathbf{x} \in \mathcal{E}_{\lambda_1}(\mathbf{A})$. The lower equality follows from

$$R_{\mathbf{A}}(\mathbf{x}) - \lambda_n = \sum_i (\lambda_i - \lambda_n) w_i.$$

□

Corollary 8.12 (Extreme eigenvalues).

$$\lambda_1(\mathbf{A}) = \max_{\|\mathbf{x}\|=1} \mathbf{x}'\mathbf{A}\mathbf{x}, \quad \lambda_n(\mathbf{A}) = \min_{\|\mathbf{x}\|=1} \mathbf{x}'\mathbf{A}\mathbf{x}.$$

The maximisers and minimisers are the unit vectors in the corresponding eigenspaces.

Origin: H Thm. 21.5.6; A&M Ex. 12.39(c)

Proof. Apply Theorem 8.11; a unit eigenvector attains each bound. $\square$

2.2 Courant–Fischer

Theorem 8.13 (Courant–Fischer). For $1 \leq k \leq n$,

$$\lambda_k = \max_{\substack{\mathcal{U} \subseteq \mathbb{R}^n \\ \dim \mathcal{U} = k}} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}}(\mathbf{x}) = \min_{\substack{\mathcal{V} \subseteq \mathbb{R}^n \\ \dim \mathcal{V} = n-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{V}} R_{\mathbf{A}}(\mathbf{x}).$$

Equivalently,

$$\lambda_k = \min_{\dim \mathcal{L} = k-1} \max_{\substack{\mathbf{0} \neq \mathbf{x} \\ \mathbf{x} \perp \mathcal{L}}} R_{\mathbf{A}}(\mathbf{x}) = \max_{\dim \mathcal{L} = n-k} \min_{\substack{\mathbf{0} \neq \mathbf{x} \\ \mathbf{x} \perp \mathcal{L}}} R_{\mathbf{A}}(\mathbf{x}).$$

Origin: S Thm. 6.2; A&M Exs. 12.42–12.44; H Ex. 21.17

Proof. Let

$$\mathcal{F}_k = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_k), \quad \mathcal{T}_k = \text{span}(\mathbf{q}_k, \dots, \mathbf{q}_n).$$

If $\dim \mathcal{U} = k$, then

$$\dim \mathcal{U} + \dim \mathcal{T}_k = n + 1 \implies \mathcal{U} \cap \mathcal{T}_k \neq \{\mathbf{0}\}.$$

For $\mathbf{0} \neq \mathbf{x} \in \mathcal{U} \cap \mathcal{T}_k$, $R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_k$. Hence

$$\min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}}(\mathbf{x}) \leq \lambda_k.$$

For $\mathcal{U} = \mathcal{F}_k$, the minimum equals λ_k , proving the first formula.

If $\dim \mathcal{V} = n - k + 1$, then $\mathcal{V} \cap \mathcal{F}_k \neq \{\mathbf{0}\}$. A nonzero vector in this intersection has Rayleigh quotient at least λ_k , so

$$\max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{V}} R_{\mathbf{A}}(\mathbf{x}) \geq \lambda_k.$$

For $\mathcal{V} = \mathcal{T}_k$, the maximum equals λ_k . This proves the second formula. Taking orthogonal complements gives the last two formulas. $\square$

Example 8.3. For $\mathbf{A}$ in Example 8.1, impose $\mathbf{x}'\mathbf{e}_3 = 0$. Writing $\mathbf{x} = (x_1, x_2, 0)'\neq \mathbf{0}$ gives

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{2x_1^2 + 2x_1x_2 + 2x_2^2}{x_1^2 + x_2^2}, \quad 1 \leq R_{\mathbf{A}}(\mathbf{x}) \leq 3.$$

The bounds are attained at $(1, -1, 0)'$ and $(1, 1, 0)'$, respectively.

Origin: Adapted from A&M Ex. 12.42

2.3 The induced norm and perturbations

Definition 8.14 (Induced Euclidean matrix norm). For $\mathbf{C} \in \mathbb{R}^{m \times n}$,

$$\|\mathbf{C}\|_2 = \sup_{\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n} \frac{\|\mathbf{C}\mathbf{x}\|}{\|\mathbf{x}\|} = \sup_{\|\mathbf{x}\|=1} \|\mathbf{C}\mathbf{x}\|.$$

For $\|\mathbf{x}\| = 1$, Cauchy–Schwarz gives

$$0 \leq \|\mathbf{C}\mathbf{x}\|^2 = \sum_{i=1}^m (\tilde{\mathbf{c}}_i' \mathbf{x})^2 \leq \sum_{i=1}^m \|\tilde{\mathbf{c}}_i\|^2,$$

so real completeness makes the supremum finite.

Origin: S §6.2, comment following Prop. 6.2

Theorem 8.15 (Norm of a symmetric matrix). If $\mathbf{A} = \mathbf{A}'$, then

$$\|\mathbf{A}\|_2 = \max_{1 \leq i \leq n} |\lambda_i(\mathbf{A})| = \max\{|\lambda_1(\mathbf{A})|, |\lambda_n(\mathbf{A})|\}.$$

The supremum in Definition 8.14 is attained by a unit eigenvector whose eigenvalue has largest modulus.

Origin: S §6.2, comment following Prop. 6.2; H §21.5f

Proof. For $\mathbf{x} = \sum_i c_i \mathbf{q}_i$,

$$\|\mathbf{Ax}\|^2 = \sum_i \lambda_i^2 c_i^2 \leq \left(\max_i |\lambda_i|\right)^2 \sum_i c_i^2 = \left(\max_i |\lambda_i|\right)^2 \|\mathbf{x}\|^2.$$

Equality holds when $\mathbf{x}$ is a corresponding unit eigenvector. □

Theorem 8.16 (Weyl perturbation bound). Let $\mathbf{A} = \mathbf{A}'$ and $\mathbf{E} = \mathbf{E}'$ be of order n . For $1 \leq k \leq n$,

$$\lambda_k(\mathbf{A}) + \lambda_n(\mathbf{E}) \leq \lambda_k(\mathbf{A} + \mathbf{E}) \leq \lambda_k(\mathbf{A}) + \lambda_1(\mathbf{E}).$$

Consequently,

$$|\lambda_k(\mathbf{A} + \mathbf{E}) - \lambda_k(\mathbf{A})| \leq \|\mathbf{E}\|_2.$$

Origin: S Thm. 6.3 and Prop. 6.2; A&M Exs. 12.40 and 12.44

Proof. For every $\mathbf{0} \neq \mathbf{x}$,

$$R_{\mathbf{A}+\mathbf{E}}(\mathbf{x}) = R_{\mathbf{A}}(\mathbf{x}) + R_{\mathbf{E}}(\mathbf{x}), \quad \lambda_n(\mathbf{E}) \leq R_{\mathbf{E}}(\mathbf{x}) \leq \lambda_1(\mathbf{E}).$$

The max–min formula gives

$$\begin{aligned} \lambda_k(\mathbf{A} + \mathbf{E}) &= \max_{\dim \mathcal{U}=k} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}+\mathbf{E}}(\mathbf{x}) \\ &\geq \max_{\dim \mathcal{U}=k} \left\{ \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}}(\mathbf{x}) + \lambda_n(\mathbf{E}) \right\} \\ &= \lambda_k(\mathbf{A}) + \lambda_n(\mathbf{E}). \end{aligned}$$

The min–max formula similarly gives the upper bound. Therefore

$$-\|\mathbf{E}\|_2 \leq \lambda_n(\mathbf{E}) \leq \lambda_k(\mathbf{A} + \mathbf{E}) - \lambda_k(\mathbf{A}) \leq \lambda_1(\mathbf{E}) \leq \|\mathbf{E}\|_2.$$

□

3 Invariant subspaces, compression, and principal components

3.1 Invariant subspaces and commuting matrices

Definition 8.17 (Invariant subspace). A subspace $\mathcal{U} \subseteq \mathbb{R}^n$ is **invariant under $\mathbf{A}$** if

$$\mathbf{A}\mathcal{U} = \{\mathbf{A}\mathbf{u} : \mathbf{u} \in \mathcal{U}\} \subseteq \mathcal{U}.$$

Origin: S §5.5.2, p. 91

Theorem 8.18 (Invariant orthogonal complement). Let $\mathbf{A} = \mathbf{A}'$ and let $\mathbf{G}$ and $\mathbf{H}$ have orthonor-

mal columns spanning $\mathcal{U}$ and $\mathcal{U}^\perp$, respectively. Then

$$\mathcal{U} \text{ is invariant under } \mathbf{A} \iff \mathcal{U}^\perp \text{ is invariant under } \mathbf{A}.$$

Under either condition,

$$(\mathbf{G}, \mathbf{H})' \mathbf{A} (\mathbf{G}, \mathbf{H}) = \text{diag}(\mathbf{G}' \mathbf{A} \mathbf{G}, \mathbf{H}' \mathbf{A} \mathbf{H}).$$

Origin: S §5.5.2, p. 91; H eqs. (21.13.3)–(21.13.4)

Proof. For $\mathbf{u} \in \mathcal{U}$ and $\mathbf{v} \in \mathcal{U}^\perp$,

$$\mathbf{v}' \mathbf{A} \mathbf{u} = (\mathbf{A} \mathbf{v})' \mathbf{u}.$$

If $\mathbf{A} \mathcal{U} \subseteq \mathcal{U}$, the left side is zero for every $\mathbf{u}$, so $\mathbf{A} \mathbf{v} \in \mathcal{U}^\perp$. The reverse implication follows by exchanging $\mathcal{U}$ and $\mathcal{U}^\perp$. In matrix form,

$$\mathbf{H}' \mathbf{A} \mathbf{G} = \mathbf{0}, \quad \mathbf{G}' \mathbf{A} \mathbf{H} = (\mathbf{H}' \mathbf{A} \mathbf{G})' = \mathbf{0},$$

which gives the block diagonal identity. □

Theorem 8.19 (Simultaneous orthogonal diagonalisation). For symmetric $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$, the following are equivalent:

$$\mathbf{AB} = \mathbf{BA};$$

$\mathbf{Q}'\mathbf{A}\mathbf{Q}$ and $\mathbf{Q}'\mathbf{B}\mathbf{Q}$ are diagonal for some orthogonal $\mathbf{Q}$.

Origin: H Cor. 21.13.2; A&M Ex. 7.51

Proof. If both transformed matrices are diagonal, then

$$\mathbf{AB} = \mathbf{Q}(\mathbf{Q}'\mathbf{A}\mathbf{Q})(\mathbf{Q}'\mathbf{B}\mathbf{Q})\mathbf{Q}' = \mathbf{Q}(\mathbf{Q}'\mathbf{B}\mathbf{Q})(\mathbf{Q}'\mathbf{A}\mathbf{Q})\mathbf{Q}' = \mathbf{BA}.$$

Conversely, suppose $\mathbf{AB} = \mathbf{BA}$. For $\mathbf{x} \in \mathcal{E}_\theta(\mathbf{A})$,

$$\mathbf{A}(\mathbf{B}\mathbf{x}) = \mathbf{BA}\mathbf{x} = \theta\mathbf{B}\mathbf{x},$$

so every eigenspace of $\mathbf{A}$ is invariant under $\mathbf{B}$. Choose an orthonormal basis matrix $\mathbf{Q}_j$ for each eigenspace. Then

$$\mathbf{B}\mathbf{Q}_j = \mathbf{Q}_j\mathbf{C}_j, \quad \mathbf{C}_j = \mathbf{Q}_j'\mathbf{B}\mathbf{Q}_j = \mathbf{C}_j'.$$

The spectral theorem gives an orthogonal $\mathbf{R}_j$ with $\mathbf{R}_j' \mathbf{C}_j \mathbf{R}_j$ diagonal. Hence

$$\mathbf{Q} = (\mathbf{Q}_1 \mathbf{R}_1, \dots, \mathbf{Q}_s \mathbf{R}_s)$$

is orthogonal and diagonalises both matrices. □

3.2 Compressions and interlacing

Definition 8.20 (Orthogonal compression). Let $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$ and $\mathbf{G} \in \mathbb{R}^{n \times r}$ satisfy $\mathbf{G}'\mathbf{G} = \mathbf{I}_r$. The matrix

$$\mathbf{C} = \mathbf{G}'\mathbf{A}\mathbf{G}$$

is the **compression of $\mathbf{A}$ to $\text{col } \mathbf{G}$** in the orthonormal basis formed by the columns of $\mathbf{G}$. It is symmetric and

$$R_{\mathbf{C}}(\mathbf{y}) = R_{\mathbf{A}}(\mathbf{G}\mathbf{y}), \quad \mathbf{0} \neq \mathbf{y} \in \mathbb{R}^r.$$

Origin: S §6.4, p. 120; A&M Ex. 12.46

Theorem 8.21 (Poincaré separation). Let

$$\lambda_1 \geq \cdots \geq \lambda_n \quad \text{and} \quad \mu_1 \geq \cdots \geq \mu_r$$

be the eigenvalues of $\mathbf{A}$ and $\mathbf{G}'\mathbf{A}\mathbf{G}$, respectively, where $\mathbf{G}'\mathbf{G} = \mathbf{I}_r$. Then

$$\lambda_k \geq \mu_k \geq \lambda_{n-r+k}, \quad 1 \leq k \leq r.$$

Origin: A&M Ex. 12.46; S Thm. 6.5

Proof. The map $\mathbf{y} \mapsto \mathbf{G}\mathbf{y}$ preserves inner products and gives a bijection between subspaces of $\mathbb{R}^r$ and subspaces of $\text{col } \mathbf{G}$ of the same dimension. Courant–Fischer therefore gives

$$\begin{aligned} \mu_k &= \max_{\substack{\mathcal{U} \subseteq \text{col } \mathbf{G} \\ \dim \mathcal{U} = k}} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}}(\mathbf{x}) \\ &\leq \max_{\substack{\mathcal{U} \subseteq \mathbb{R}^n \\ \dim \mathcal{U} = k}} \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{U}} R_{\mathbf{A}}(\mathbf{x}) = \lambda_k, \end{aligned}$$

and

$$\begin{aligned}\mu_k &= \min_{\substack{\mathcal{V} \subseteq \text{col } \mathbf{G} \\ \dim \mathcal{V} = r-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{V}} R_{\mathbf{A}}(\mathbf{x}) \\ &\geq \min_{\substack{\mathcal{V} \subseteq \mathbb{R}^n \\ \dim \mathcal{V} = r-k+1}} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{V}} R_{\mathbf{A}}(\mathbf{x}) = \lambda_{n-r+k}.\end{aligned}$$

□

Corollary 8.22 (Cauchy interlacing). Let $n \geq 2$, and let $\mathbf{C}$ be an $(n-1) \times (n-1)$ principal submatrix of a real symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$. If

$$\lambda_1 \geq \cdots \geq \lambda_n, \quad \mu_1 \geq \cdots \geq \mu_{n-1}$$

are the respective eigenvalues, then

$$\lambda_k \geq \mu_k \geq \lambda_{k+1}, \quad 1 \leq k \leq n-1.$$

Origin: S Thm. 6.5; A&M Ex. 12.48

Proof. If $i_1, \dots, i_{n-1}$ are the retained indices, set $\mathbf{G} = (\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_{n-1}})$. Then

$$\mathbf{G}'\mathbf{G} = \mathbf{I}_{n-1}, \quad \mathbf{C} = \mathbf{G}'\mathbf{A}\mathbf{G}.$$

Apply Theorem 8.21 with $r = n - 1$. □

3.3 Principal components and Ky Fan's principle

Definition 8.23 (Sample principal components). Let $\mathbf{X} \in \mathbb{R}^{m \times p}$ contain $m \geq 2$ observations by rows, and set

$$\mathbf{Z} = \left(\mathbf{I}_m - \frac{1}{m} \mathbf{1} \mathbf{1}' \right) \mathbf{X}, \quad \mathbf{S} = \frac{1}{m-1} \mathbf{Z}' \mathbf{Z}.$$

For $\mathbf{a} \in \mathbb{R}^p$ with $\|\mathbf{a}\| = 1$, $\mathbf{Za}$ is a **principal-component score candidate**; its sample variance is

$$\frac{1}{m-1} \|\mathbf{Za}\|^2 = \mathbf{a}' \mathbf{S} \mathbf{a}.$$

Successive **principal-component loading vectors** maximise this variance subject to being unit and orthogonal to the preceding loading vectors.

Origin: Adapted from A&M Exs. 12.39 and 12.49

Theorem 8.24 (Ky Fan trace principle). Let $\mathbf{S} = \mathbf{S}' \in \mathbb{R}^{p \times p}$ and

$$\mathbf{S} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_p) \mathbf{Q}', \quad \lambda_1 \geq \dots \geq \lambda_p.$$

For $1 \leq r \leq p$,

$$\max_{\substack{\mathbf{G} \in \mathbb{R}^{p \times r} \\ \mathbf{G}'\mathbf{G} = \mathbf{I}_r}} \operatorname{tr}(\mathbf{G}'\mathbf{S}\mathbf{G}) = \sum_{i=1}^r \lambda_i.$$

If $r < p$ and $\lambda_r > \lambda_{r+1}$, equality in this maximisation holds exactly when

$$\operatorname{col} \mathbf{G} = \operatorname{span}(\mathbf{q}_1, \dots, \mathbf{q}_r).$$

Origin: A&M Ex. 12.49; S Thm. 6.2

Proof. Let $\mathbf{C} = \mathbf{Q}'\mathbf{G} = (c_{ij})$ and

$$w_i = \sum_{j=1}^r c_{ij}^2.$$

Since $\mathbf{C}'\mathbf{C} = \mathbf{I}_r$,

$$0 \leq w_i = \|\mathbf{G}'\mathbf{q}_i\|^2, \quad 1 - w_i = \|(\mathbf{I} - \mathbf{G}\mathbf{G}')\mathbf{q}_i\|^2 \geq 0,$$

and

$$\sum_{i=1}^p w_i = r, \quad \text{tr}(\mathbf{G}'\mathbf{S}\mathbf{G}) = \sum_{i=1}^p \lambda_i w_i.$$

For $\alpha = \lambda_r$,

$$\begin{aligned} \sum_{i=1}^p \lambda_i w_i &= \alpha r + \sum_{i < r} (\lambda_i - \alpha) w_i + \sum_{i > r} (\lambda_i - \alpha) w_i \\ &\leq \alpha r + \sum_{i < r} (\lambda_i - \alpha) = \sum_{i=1}^r \lambda_i. \end{aligned}$$

The bound is attained by $\mathbf{G} = (\mathbf{q}_1, \dots, \mathbf{q}_r)$. If $\lambda_r > \lambda_{r+1}$, equality forces $w_i = 0$ for $i > r$. Then $\sum_{i=1}^r w_i = r$ and $w_i \leq 1$ force $w_i = 1$ for $i \leq r$, which is equivalent to the stated column space. $\square$

Corollary 8.25 (Principal-component solution). Let $\mathbf{Z}$ and $\mathbf{S} = \mathbf{Z}'\mathbf{Z}/(m-1)$ be as in Definition 8.23, and write

$$\mathbf{S} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_p) \mathbf{Q}', \quad \lambda_1 \geq \dots \geq \lambda_p.$$

Then $\lambda_p \geq 0$. The successive principal-component loadings may be chosen as $\mathbf{q}_1, \dots, \mathbf{q}_p$, with component variances $\lambda_1, \dots, \lambda_p$. For $1 \leq r \leq p$,

$$\min_{\mathbf{G}'\mathbf{G}=\mathbf{I}_r} \|\mathbf{Z} - \mathbf{Z}\mathbf{G}\mathbf{G}'\|_F^2 = (m-1) \sum_{i=r+1}^p \lambda_i.$$

The sum is 0 when $r = p$. If $r < p$ and $\lambda_r > \lambda_{r+1}$, the minimising column space is uniquely $\operatorname{span}(\mathbf{q}_1, \dots, \mathbf{q}_r)$.

Origin: Adapted from A&M Exs. 12.39 and 12.49

Proof. For every $\mathbf{a}$,

$$\mathbf{a}'\mathbf{S}\mathbf{a} = \frac{\|\mathbf{Z}\mathbf{a}\|^2}{m-1} \geq 0.$$

Theorem 8.11 gives $\lambda_p \geq 0$. For $1 \leq k \leq p$, if $\mathbf{a} \perp \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_{k-1})$ and $\|\mathbf{a}\| = 1$, then

$$\mathbf{a} = \sum_{i=k}^p c_i \mathbf{q}_i, \quad \mathbf{a}'\mathbf{S}\mathbf{a} = \sum_{i=k}^p \lambda_i c_i^2 \leq \lambda_k, \quad \sum_{i=k}^p c_i^2 = 1.$$

The choice $\mathbf{a} = \mathbf{q}_k$ attains the bound, proving the successive loading assertions.

For $\mathbf{P} = \mathbf{G}\mathbf{G}'$, $\mathbf{P}' = \mathbf{P}^2 = \mathbf{P}$, so cyclicity of trace gives

$$\begin{aligned} \|\mathbf{Z} - \mathbf{Z}\mathbf{P}\|_F^2 &= \text{tr}\{(\mathbf{I} - \mathbf{P})\mathbf{Z}'\mathbf{Z}(\mathbf{I} - \mathbf{P})\} \\ &= \text{tr}(\mathbf{Z}'\mathbf{Z}) - \text{tr}(\mathbf{G}'\mathbf{Z}'\mathbf{Z}\mathbf{G}) \\ &= (m - 1)\{\text{tr} \mathbf{S} - \text{tr}(\mathbf{G}'\mathbf{S}\mathbf{G})\}. \end{aligned}$$

Theorem 8.24 and $\text{tr} \mathbf{S} = \sum_i \lambda_i$ give the minimum. Its equality case gives uniqueness under the gap. □

Example 8.4. For

$$\mathbf{Z} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{S} = \frac{1}{2}\mathbf{Z}'\mathbf{Z} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

the loadings

$$\mathbf{q}_1 = 2^{-1/2}(1, 1)', \quad \mathbf{q}_2 = 2^{-1/2}(1, -1)'$$

have variances 2 and 0. Also

$$\mathbf{Z}\mathbf{q}_1\mathbf{q}_1' = \mathbf{Z}, \quad \|\mathbf{Z} - \mathbf{Z}\mathbf{q}_1\mathbf{q}_1'\|_F^2 = 0.$$

Origin: Adapted from A&M Ex. 12.49

4 Computation: Numerical Checks in R

4.1 Spectral factor and projectors (§1)

```
1 A <- matrix(c(4, 1, 0, 1,  
2               1, 3, 1, 0,  
3               0, 1, 2, 1,
```

```

4           1, 0, 1, 3), 4, 4, byrow = TRUE)
5 ea <- eigen(A, symmetric = TRUE)
6 Q <- ea$vectors
7 Lam <- diag(ea$values)
8 c(orthogonality = max(abs(crossprod(Q) - diag(4))),
9   reconstruction = max(abs(A - Q %*% Lam %*% t(Q))))

```

```

## orthogonality reconstruction
##      3.331e-16      6.661e-15

```

```

1 M <- matrix(c(1, 1, 0, 0,
2               1, 0, 1, 0,
3               0, 1, 1, 1,
4               0, 0, 1, 1), 4, 4, byrow = TRUE)
5 Q0 <- qr.Q(qr(M))
6 Arep <- Q0 %*% diag(c(4, 4, 1, -2)) %*% t(Q0)
7 P_basis <- Q0[, 1:2] %*% t(Q0[, 1:2])
8 P_poly <- (Arep - diag(4)) %*% (Arep + 2 * diag(4)) / 18
9 c(projector_routes = max(abs(P_basis - P_poly)),
10   idempotence = max(abs(P_poly %*% P_poly - P_poly)),
11   symmetry = max(abs(P_poly - t(P_poly))))

```

```
## projector_routes      idempotence      symmetry
##          3.331e-16          3.331e-16      2.467e-17
```

4.2 Rayleigh bounds and the induced norm (§2.1)

```
1 set.seed(802)
2 X <- matrix(rnorm(4000), 4, 1000)
3 X <- sweep(X, 2, sqrt(colSums(X^2)), "/")
4 rq <- colSums(X * (A %*% X))
5 manual_norm <- max(abs(ea$values))
6 package_norm <- norm(A, type = "2")
7 c(sample_min_minus_lambda_min = min(rq) - min(ea$values),
8   lambda_max_minus_sample_max = max(ea$values) - max(rq),
9   norm_difference = manual_norm - package_norm)
```

```
## sample_min_minus_lambda_min lambda_max_minus_sample_max
##          0.06411          0.04380
##          norm_difference
##          0.00000
```

4.3 Weyl's perturbation bound (§2.3)

```

1 E <- matrix(c(0.10, -0.04, 0, 0,
2               -0.04, -0.08, 0.03, 0,
3               0, 0.03, 0.05, -0.02,
4               0, 0, -0.02, -0.01), 4, 4, byrow = TRUE)
5 lam_A <- eigen(A, symmetric = TRUE, only.values = TRUE)$values
6 lam_AE <- eigen(A + E, symmetric = TRUE, only.values = TRUE)$values
7 bound <- max(abs(eigen(E, symmetric = TRUE, only.values = TRUE)$values))
8 c(max_eigenvalue_movement = max(abs(lam_AE - lam_A)),
9   induced_norm_bound = bound,
10  bound_slack = bound - max(abs(lam_AE - lam_A)))

```

```

## max_eigenvalue_movement      induced_norm_bound
##                0.07494                0.10924
##                bound_slack
##                0.03431

```

4.4 Simultaneous diagonalisation (§3.1)

```

1 Q1 <- qr.Q(qr(matrix(c(1, 2, 0, 1,
2                       0, 1, 2, 1,

```

```

3           1, 0, 1, 2,
4           2, 1, 1, 0), 4, 4, byrow = TRUE)))
5 As <- Q1 %%% diag(c(3, 3, 1, 1)) %%% t(Q1)
6 Bs <- Q1 %%% diag(c(4, -1, 2, 5)) %%% t(Q1)
7 va <- eigen(As, symmetric = TRUE)$vectors
8 groups <- split(seq_len(4), round(diag(crossprod(va, As %%% va)), 8))
9 Qsim <- matrix(0, 4, 4)
10 for (idx in groups) {
11   Vg <- va[, idx, drop = FALSE]
12   Rg <- eigen(crossprod(Vg, Bs %%% Vg), symmetric = TRUE)$vectors
13   Qsim[, idx] <- Vg %%% Rg
14 }
15 offdiag_max <- function(M) max(abs(M - diag(diag(M))))
16 c(commutator = max(abs(As %%% Bs - Bs %%% As)),
17   A_off_diagonal = offdiag_max(crossprod(Qsim, As %%% Qsim)),
18   B_off_diagonal = offdiag_max(crossprod(Qsim, Bs %%% Qsim)))

```

```

##      commutator A_off_diagonal B_off_diagonal
##      2.665e-15      1.866e-15      5.655e-16

```

4.5 Compression and interlacing (§3.2)

```

1 G <- qr.Q(qr(matrix(c(1, 0, 1,
2                       0, 1, 1,
3                       1, 1, 0,
4                       1, -1, 1), 4, 3, byrow = TRUE)))
5 mu <- eigen(crossprod(G, A %*% G), symmetric = TRUE,
6             only.values = TRUE)$values
7 lambda <- ea$values
8 interlace <- cbind(upper = lambda[1:3], compression = mu,
9                   lower = lambda[2:4])
10 interlace

```

```

##      upper compression lower
## [1,] 5.236          5.187 3.0000
## [2,] 3.000          3.000 3.0000
## [3,] 3.000          1.080 0.7639

```

```

1 c(upper_slack_min = min(interlace[, "upper"] - interlace[, "compression"]),
2   lower_slack_min = min(interlace[, "compression"] - interlace[, "lower"]))

```

```
## upper_slack_min lower_slack_min
```

```
##          2.665e-15          8.882e-16
```

4.6 Principal components from a sample covariance (§3.3)

```
1 data("Boston", package = "ISLR2")
2 Z <- scale(as.matrix(Boston), center = TRUE, scale = FALSE)
3 S <- crossprod(Z) / (nrow(Z) - 1)
4 es <- eigen(S, symmetric = TRUE)
5 rotated_cov <- crossprod(Z %*% es$vectors) / (nrow(Z) - 1)
6 ratio <- es$values / sum(es$values)
7 reconstruction <-
8   norm(Z - Z %*% es$vectors[, 1, drop = FALSE] %*%
9     t(es$vectors[, 1, drop = FALSE]), type = "F")^2
10 spectral_tail <- (nrow(Z) - 1) * sum(es$values[-1])
11 c(rotated_off_diagonal = offdiag_max(rotated_cov),
12   variance_ratio_sum_minus_one = sum(ratio) - 1,
13   one_component_reconstruction = reconstruction,
14   spectral_tail_formula = spectral_tail,
15   reconstruction_minus_tail = reconstruction - spectral_tail)
```

```
##          rotated_off_diagonal variance_ratio_sum_minus_one
##          5.187e-12          0.000e+00
```

```
## one_component_reconstruction      spectral_tail_formula
##                                6.385e+05      6.385e+05
## reconstruction_minus_tail
##                                -2.095e-09
```

5 Pitfalls

5.1 Comparing eigenvectors coordinatewise

$$\mathbf{q} \quad \text{and} \quad -\mathbf{q}$$

represent the same one-dimensional eigenspace. For simple eigenvalues, compare $|\mathbf{q}'\tilde{\mathbf{q}}|$ or $\mathbf{q}\mathbf{q}'$ with $\tilde{\mathbf{q}}\tilde{\mathbf{q}}'$. For a repeated eigenvalue, compare its full spectral projector.

5.2 Replacing the transpose by an unrelated inverse

$$\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1} \not\Rightarrow \mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}'.$$

The second identity requires $\mathbf{S}'\mathbf{S} = \mathbf{I}$. A real matrix admits such a diagonalisation exactly when it is symmetric.

5.3 Treating sampled Rayleigh quotients as extrema

$$\max_{1 \leq j \leq N} R_{\mathbf{A}}(\mathbf{x}_j) \leq \lambda_1(\mathbf{A}), \quad \min_{1 \leq j \leq N} R_{\mathbf{A}}(\mathbf{x}_j) \geq \lambda_n(\mathbf{A}).$$

Finite random searches generally do not contain an extreme eigenvector.

5.4 Confusing covariance and correlation components

If every $s_{ii} > 0$ and $\mathbf{D} = \text{diag}(s_{11}^{1/2}, \dots, s_{pp}^{1/2})$, then

$$\mathbf{S} \quad \text{and} \quad \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1}$$

usually have different eigenvectors. The first retains measurement units; the second standardises every coordinate to diagonal entry 1.

6 Exercises

Exercise 8.1. Let $\mathbf{P}$ be a spectral projector of a real symmetric matrix. Prove that

$$\mathbf{H} = \mathbf{I} - 2\mathbf{P}$$

is symmetric and orthogonal, acts as $-\mathbf{I}$ on $\text{im } \mathbf{P}$, and acts as $\mathbf{I}$ on $(\text{im } \mathbf{P})^\perp$.

Origin: H Ex. 10.3 and Ex. 21.18

Solution. Since $\mathbf{P}' = \mathbf{P}^2 = \mathbf{P}$,

$$\mathbf{H}' = \mathbf{H}, \quad \mathbf{H}'\mathbf{H} = (\mathbf{I} - 2\mathbf{P})^2 = \mathbf{I} - 4\mathbf{P} + 4\mathbf{P}^2 = \mathbf{I}.$$

If $\mathbf{x} = \mathbf{P}\mathbf{u} \in \text{im } \mathbf{P}$, then

$$\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{P}^2\mathbf{u} = -\mathbf{x}.$$

If $\mathbf{x} \perp \text{im } \mathbf{P}$, the orthogonal-projector identity gives $\mathbf{P}\mathbf{x} = \mathbf{0}$, so $\mathbf{H}\mathbf{x} = \mathbf{x}$. 

Exercise 8.2. Let $\mathbf{A} = \mathbf{A}'$ have distinct eigenvalues $\theta_1, \dots, \theta_s$, and let $\emptyset \neq J \subseteq \{1, \dots, s\}$. Construct a polynomial p of degree at most $s - 1$ such that

$$p(\mathbf{A})$$

is the orthogonal projector onto $\perp \oplus_{j \in J} \mathcal{E}_{\theta_j}(\mathbf{A})$.

Origin: S §5.5; H Ex. 21.18

Solution. With the Lagrange polynomials from Theorem 8.6, set

$$p(t) = \sum_{j \in J} \ell_j(t).$$

Then $\deg p \leq s - 1$ and

$$p(\theta_i) = \begin{cases} 1, & i \in J, \\ 0, & i \notin J. \end{cases}$$

Hence

$$p(\mathbf{A}) = \sum_{i=1}^s p(\theta_i) \mathbf{P}_i = \sum_{i \in J} \mathbf{P}_i,$$

which is the stated orthogonal projector. 

Exercise 8.3. Let $\mathbf{A} = \mathbf{A}'$. Prove that $\mathbf{A}^k$ converges entrywise as $k \rightarrow \infty$ if and only if every eigenvalue lies in $(-1, 1]$. When it converges, identify the limit.

Origin: Adapted from H Ex. 21.19

Solution. Finite spectral calculus gives

$$\mathbf{A}^k = \sum_j \theta_j^k \mathbf{P}_j.$$

If every $\theta_j \in (-1, 1]$, then

$$\theta_j^k \longrightarrow \begin{cases} 1, & \theta_j = 1, \\ 0, & -1 < \theta_j < 1, \end{cases}$$

so $\mathbf{A}^k$ converges to the spectral projector for the eigenvalue 1, or to $\mathbf{O}$ if 1 is absent.

Conversely, if $\mathbf{A}^k$ converges, then for a unit eigenvector $\mathbf{q}_j$,

$$\mathbf{q}'_j \mathbf{A}^k \mathbf{q}_j = \theta_j^k$$

converges. A real geometric sequence converges exactly when its ratio lies in $(-1, 1]$. 

Exercise 8.4. Let $\mathbf{A} = \mathbf{A}'$ and let p be a real polynomial. Prove

$$\|p(\mathbf{A})\|_2 = \max_i |p(\lambda_i(\mathbf{A}))|.$$

Deduce that

$$\|\mathbf{A}^k\|_2 = \|\mathbf{A}\|_2^k, \quad k \geq 1.$$

Origin: S Prop. 5.7 and §6.2, comment following Prop. 6.2

Solution. The eigenvalues of the symmetric matrix $p(\mathbf{A})$ are $p(\lambda_i(\mathbf{A}))$. Theorem 8.15 gives the first identity. With $p(t) = t^k$,

$$\|\mathbf{A}^k\|_2 = \max_i |\lambda_i|^k = \left(\max_i |\lambda_i| \right)^k = \|\mathbf{A}\|_2^k. \quad \text{■}$$

Exercise 8.5. Let $\mathbf{A} = \mathbf{A}'$, let $\|\mathbf{x}\| = 1$, and set

$$\mu = \mathbf{x}'\mathbf{A}\mathbf{x}, \quad \mathbf{r} = \mathbf{A}\mathbf{x} - \mu\mathbf{x}.$$

Prove

$$\mathbf{x}'\mathbf{r} = 0, \quad \min_i |\lambda_i(\mathbf{A}) - \mu| \leq \|\mathbf{r}\|.$$

Origin: Adapted from S Thm. 6.2

Solution.

$$\mathbf{x}'\mathbf{r} = \mathbf{x}'\mathbf{A}\mathbf{x} - \mu\mathbf{x}'\mathbf{x} = 0.$$

Write $\mathbf{x} = \sum_i c_i \mathbf{q}_i$. Since $\sum_i c_i^2 = 1$,

$$\|\mathbf{r}\|^2 = \sum_i c_i^2 (\lambda_i - \mu)^2 \geq \left(\min_i |\lambda_i - \mu| \right)^2 \sum_i c_i^2.$$

Both sides are nonnegative, so the bound follows. 

Exercise 8.6. Let $\mathbf{A} = \mathbf{A}'$ and $\mathbf{u} \in \mathbb{R}^n$. Prove

$$0 \leq \lambda_k(\mathbf{A} + \mathbf{u}\mathbf{u}') - \lambda_k(\mathbf{A}) \leq \|\mathbf{u}\|^2, \quad 1 \leq k \leq n.$$

Origin: Adapted from S Ex. 6.4 and Prop. 6.2; A&M Ex. 12.40

Solution. For every unit $\mathbf{x}$,

$$0 \leq \mathbf{x}'\mathbf{u}\mathbf{u}'\mathbf{x} = (\mathbf{u}'\mathbf{x})^2 \leq \|\mathbf{u}\|^2.$$

Hence the smallest and largest eigenvalues of $\mathbf{u}\mathbf{u}'$ are 0 and $\|\mathbf{u}\|^2$ when $n \geq 2$; for $n = 1$ both equal $\|\mathbf{u}\|^2$. Weyl's bounds give, in either case,

$$0 \leq \lambda_k(\mathbf{A} + \mathbf{u}\mathbf{u}') - \lambda_k(\mathbf{A}) \leq \|\mathbf{u}\mathbf{u}'\|_2 = \|\mathbf{u}\|^2.$$



Exercise 8.7. Let $\mathbf{A}_1, \dots, \mathbf{A}_m$ be real symmetric matrices that commute in pairs. Prove that one orthogonal matrix simultaneously diagonalises all of them.

Origin: Adapted from H Thm. 21.13.1 and Ex. 21.29

Solution. Induct on m . The assertion for $m = 1$ is the spectral theorem. Suppose it holds for $m - 1$. Each eigenspace $\mathcal{E}$ of $\mathbf{A}_m$ is invariant under every $\mathbf{A}_i$, $i < m$, because

$$\mathbf{A}_m \mathbf{A}_i \mathbf{x} = \mathbf{A}_i \mathbf{A}_m \mathbf{x} = \theta \mathbf{A}_i \mathbf{x} \quad (\mathbf{x} \in \mathcal{E}).$$

The restrictions of $\mathbf{A}_1, \dots, \mathbf{A}_{m-1}$ to $\mathcal{E}$ are symmetric and commute in pairs. By induction they have a common orthonormal eigenbasis of $\mathcal{E}$. Combining these bases over the mutually orthogonal eigenspaces of $\mathbf{A}_m$ gives the required basis of $\mathbb{R}^n$. ■

Exercise 8.8. Let $\mathbf{A} = \mathbf{A}'$, $\mathbf{G}'\mathbf{G} = \mathbf{I}_r$, and suppose $\text{col } \mathbf{G}$ is invariant under $\mathbf{A}$. Prove that every eigenvalue of $\mathbf{G}'\mathbf{A}\mathbf{G}$ is an eigenvalue of $\mathbf{A}$, with no larger multiplicity.

Origin: S §6.4; H eqs. (21.13.3)–(21.13.4)

Solution. Complete $\mathbf{G}$ to an orthogonal matrix $(\mathbf{G}, \mathbf{H})$. By Theorem 8.18,

$$(\mathbf{G}, \mathbf{H})' \mathbf{A} (\mathbf{G}, \mathbf{H}) = \text{diag}(\mathbf{G}'\mathbf{A}\mathbf{G}, \mathbf{H}'\mathbf{A}\mathbf{H}).$$

Therefore

$$\chi_{\mathbf{A}}(t) = \chi_{\mathbf{G}'\mathbf{A}\mathbf{G}}(t) \chi_{\mathbf{H}'\mathbf{A}\mathbf{H}}(t).$$

Every root of the first factor is a root of the product, and its exponent in the first factor cannot exceed its exponent in the product. ■

Exercise 8.9. Let $w_{ij} = w_{ji} \geq 0$ and $w_{ii} = 0$ for $1 \leq i, j \leq n$. Define

$$\mathbf{L} = (\ell_{ij}), \quad \ell_{ii} = \sum_{j=1}^n w_{ij}, \quad \ell_{ij} = -w_{ij} \quad (i \neq j).$$

Prove

$$\mathbf{x}'\mathbf{L}\mathbf{x} = \sum_{1 \leq i < j \leq n} w_{ij}(x_i - x_j)^2.$$

Deduce that every eigenvalue of $\mathbf{L}$ is nonnegative and that $\mathbf{1}$ is an eigenvector with eigenvalue 0.

Origin: Course formulation

Solution. Symmetry and $w_{ii} = 0$ give

$$\begin{aligned}\mathbf{x}'\mathbf{L}\mathbf{x} &= \sum_i \left(\sum_j w_{ij} \right) x_i^2 - 2 \sum_{i < j} w_{ij} x_i x_j \\ &= \sum_{i < j} w_{ij} (x_i^2 - 2x_i x_j + x_j^2).\end{aligned}$$

For a unit eigenvector $\mathbf{q}$ with eigenvalue λ ,

$$\lambda = \mathbf{q}'\mathbf{L}\mathbf{q} \geq 0.$$

Finally,

$$(\mathbf{L}\mathbf{1})_i = \sum_j w_{ij} - \sum_{j \neq i} w_{ij} = 0.$$



Exercise 8.10. Let $d_1 \geq \cdots \geq d_n$ be the diagonal entries of a symmetric $\mathbf{A}$, rearranged in decreasing order, and let $\lambda_1 \geq \cdots \geq \lambda_n$ be its eigenvalues. Prove

$$\sum_{i=1}^k d_i \leq \sum_{i=1}^k \lambda_i \quad (1 \leq k < n), \quad \sum_{i=1}^n d_i = \sum_{i=1}^n \lambda_i.$$

Origin: S Thm. 6.7; A&M Ex. 12.49

Solution. Let $i_1, \dots, i_k$ index the k largest diagonal entries and set $\mathbf{G} = (\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k})$. Then $\mathbf{G}'\mathbf{G} = \mathbf{I}_k$, and Ky Fan's principle gives

$$\sum_{j=1}^k d_j = \text{tr}(\mathbf{G}'\mathbf{A}\mathbf{G}) \leq \sum_{j=1}^k \lambda_j.$$

For $k = n$ both sums equal $\text{tr } \mathbf{A}$. ■

Exercise 8.11. Let $\mathbf{Z} \in \mathbb{R}^{m \times p}$ have zero column means and let $\mathbf{Q} \in \mathbb{R}^{p \times p}$ be orthogonal. Prove that the sample covariance of $\mathbf{ZQ}$ is $\mathbf{Q}'\mathbf{SQ}$, where $\mathbf{S} = \mathbf{Z}'\mathbf{Z}/(m-1)$. Deduce that total sample variance and the ordered principal-component variances are unchanged by the coordinate rotation.

Origin: Adapted from H Thm. 21.6.1 and A&M Ex. 12.49

Solution. Since $\mathbf{1}'\mathbf{Z} = \mathbf{0}'$, also $\mathbf{1}'\mathbf{ZQ} = \mathbf{0}'$. Its sample covariance is

$$\frac{1}{m-1}(\mathbf{ZQ})'(\mathbf{ZQ}) = \mathbf{Q}'\mathbf{SQ}.$$

Similarity preserves the eigenvalues, so the ordered component variances are unchanged. Also

$$\text{tr}(\mathbf{Q}'\mathbf{SQ}) = \text{tr}(\mathbf{SQQ}') = \text{tr} \mathbf{S}.$$



Exercise 8.12. For the centred matrix

$$\mathbf{Z} = \left(\mathbf{I}_m - \frac{1}{m} \mathbf{1}\mathbf{1}' \right) \mathbf{X},$$

prove

$$\text{rank } \mathbf{Z} \leq \min\{p, m - 1\}, \quad \text{rank}(\mathbf{Z}'\mathbf{Z}) = \text{rank } \mathbf{Z}.$$

Deduce that at most $\min\{p, m - 1\}$ sample principal-component variances are nonzero.

Origin: A&M Exs. 7.49–7.50; H §21.5f

Solution. The identity $\mathbf{1}'\mathbf{Z} = \mathbf{0}'$ makes every column of $\mathbf{Z}$ lie in the $(m - 1)$ -dimensional space $\mathbf{1}^\perp$. Hence

$$\text{rank } \mathbf{Z} \leq m - 1, \quad \text{rank } \mathbf{Z} \leq p.$$

Moreover,

$$\mathbf{Z}'\mathbf{Z}\mathbf{x} = \mathbf{0} \iff 0 = \mathbf{x}'\mathbf{Z}'\mathbf{Z}\mathbf{x} = \|\mathbf{Z}\mathbf{x}\|^2 \iff \mathbf{Z}\mathbf{x} = \mathbf{0}.$$

Thus the two kernels have the same dimension, and rank–nullity gives $\text{rank}(\mathbf{Z}'\mathbf{Z}) = \text{rank } \mathbf{Z}$. The rank formula in Corollary 8.7 gives the last assertion. ■

Exercise 8.13. Let $\mathbf{S} = \mathbf{S}'$ have $\lambda_1 \geq \cdots \geq \lambda_p$, and let $\mathbf{G}'\mathbf{G} = \mathbf{I}_r$, $r < p$. Assume $\lambda_r > \lambda_{r+1}$.

Prove that

$$\operatorname{tr}(\mathbf{G}'\mathbf{S}\mathbf{G}) = \sum_{i=1}^r \lambda_i$$

if and only if $\operatorname{col} \mathbf{G} = \operatorname{span}(\mathbf{q}_1, \dots, \mathbf{q}_r)$ for an ordered orthonormal eigenbasis.

Origin: Adapted from A&M Ex. 12.49

Solution. Use the notation $\mathbf{C} = \mathbf{Q}'\mathbf{G}$ and $w_i = \sum_j c_{ij}^2$ from Theorem 8.24. Its proof gives

$$0 \leq w_i \leq 1, \quad \sum_i w_i = r, \quad \sum_i \lambda_i w_i \leq \sum_{i=1}^r \lambda_i.$$

Because $\lambda_r > \lambda_{r+1}$, equality forces

$$w_i = 1 \ (i \leq r), \quad w_i = 0 \ (i > r).$$

These identities hold exactly when every column of $\mathbf{G}$ lies in $\operatorname{span}(\mathbf{q}_1, \dots, \mathbf{q}_r)$; both spaces have dimension r , so they are equal. The converse follows by substituting any orthonormal basis of that space. 

Exercise 8.14. Let $1 \leq r \leq n$, let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be orthonormal vectors in $\mathbb{R}^n$, and let $\alpha_1, \dots, \alpha_r, c \in \mathbb{R}$. Construct a symmetric matrix $\mathbf{A}$ such that

$$\mathbf{A}\mathbf{u}_i = \alpha_i \mathbf{u}_i \quad (1 \leq i \leq r), \quad \mathbf{A}\mathbf{v} = c\mathbf{v} \quad \text{for every } \mathbf{v} \perp \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r).$$

Prove uniqueness.

Origin: Adapted from H eq. (21.5.5)

Solution. Let $\mathbf{P}_i = \mathbf{u}_i \mathbf{u}_i'$ and

$$\mathbf{P}_0 = \mathbf{I} - \sum_{i=1}^r \mathbf{P}_i.$$

Then

$$\mathbf{A} = \sum_{i=1}^r \alpha_i \mathbf{P}_i + c\mathbf{P}_0$$

is symmetric and has the required action. If $\tilde{\mathbf{A}}$ has the same action, then $\mathbf{A}$ and $\tilde{\mathbf{A}}$ agree on the orthogonal basis formed by $\mathbf{u}_1, \dots, \mathbf{u}_r$ and any orthonormal basis of their orthogonal complement. Hence $\mathbf{A} = \tilde{\mathbf{A}}$. 

Exercise 8.15. For an arbitrary real square $\mathbf{A}$, set

$$\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}'), \quad \mathbf{K} = \frac{1}{2}(\mathbf{A} - \mathbf{A}').$$

Prove

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \mathbf{x}'\mathbf{H}\mathbf{x} \quad \text{for every } \mathbf{x}.$$

Give two real 2×2 matrices with the same scalar expression $\mathbf{x}'\mathbf{A}\mathbf{x}$ but different eigenvalues.

Origin: Adapted from S §5.6

Solution. Since $\mathbf{K}' = -\mathbf{K}$, the scalar $s = \mathbf{x}'\mathbf{K}\mathbf{x}$ satisfies

$$s = s' = \mathbf{x}'\mathbf{K}'\mathbf{x} = -\mathbf{x}'\mathbf{K}\mathbf{x} = -s,$$

so $s = 0$ and $\mathbf{x}'\mathbf{A}\mathbf{x} = \mathbf{x}'\mathbf{H}\mathbf{x}$.

Take

$$\mathbf{A}_1 = \mathbf{O}_2, \quad \mathbf{A}_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both scalar expressions vanish for every real $\mathbf{x}$, whereas

$$\chi_{\mathbf{A}_1}(t) = t^2, \quad \chi_{\mathbf{A}_2}(t) = t^2 + 1.$$

Thus their complex eigenvalues are $0, 0$ and $i, -i$. 